

Schwann_NF2_Normal_Tumor_Comparison copy

September 16, 2026

```
[1]: import numpy as np
import scanpy as sc
import pandas as pd
import matplotlib.pyplot as plt
import anndata as ad
```

```
[37]: color_dict = {
    'NF2_depleted': '#1f77b4',
    'NF2_indeterminate': '#ff7f0e',
    'NF2_intact': '#2ca02c',
    'Schwann_Stress': '#34495e',
    'Schwann_Macrophage': '#3498db',
    'Schwann_Classical': '#9b59b6',
    'Schwann_IFN_Response': '#2ecc71',
    'Schwann_Endothelial-like': '#e74c3c',
    'Schwann_Hypoxia': '#f39c12',
    'Schwann_Repair-like': '#1abc9c',
    'Schwann_Proliferating': '#27ae60',
    'Schwann_Myelinating': '#c0392b',
    'Macrophage_M1-like': '#2e89cb', # Adjusted
    'Macrophage_M2-like': '#f4403a', # Adjusted
    'Macrophage_Pro-Inflammatory': '#fb8309', # Adjusted
    'Macrophage_Regulatory': '#926bb1', # Adjusted
    'Macrophage_Proliferating': '#1ab15f', # Adjusted
    'Macrophage': '#3498db',
    'Endothelial': '#e74c3c',
    'T-cells': '#2ecc71',
    'Fibroblast': '#f39c12',
    'Unknown': '#1abc9c',
    'Schwann': '#9b59b6',
    'Pericytes': '#e67e22',
    'Neutrophil': '#27ae60',
    'NK Cells': '#c0392b',
    'Schwann_NF2_depleted': '#1f77b4',
    'Schwann_NF2_indeterminate': '#ff7f0e',
    'Schwann_NF2_intact': '#2ca02c',
    'Tumor': '#E41A1C',
    'Normal_Schwann_Donor': '#66C2A5',
```

```

    "Normal_Schwann_Nerve": "#8DA0CB",
    "Normal_Schwann_Spine": "#FFC0CB"
}

```

```

[6]: adata_schwann_donor = sc.read_h5ad("/data/NUPID/santosh_kumar/CNS_tumors/
    ↪lab_schwannoma/schw_adata/donor_schwann.h5ad")
adata_schwann_donor.obs['condition'] = "Normal_Schwann_Donor"

```

```

[7]: adata_schwann_nerve = sc.read_h5ad("/data/NUPID/santosh_kumar/CNS_tumors/
    ↪lab_schwannoma/schw_adata/nerve_schwann.h5ad")
adata_schwann_nerve.obs['condition'] = "Normal_Schwann_Nerve"

```

```

[8]: adata_schwann_spinal = sc.read_h5ad("/data/NUPID/santosh_kumar/CNS_tumors/
    ↪lab_schwannoma/schw_adata/spinal_schwann.h5ad")
adata_schwann_spinal.obs['condition'] = "Normal_Schwann_Spine"

```

```

[10]: adata_tumor = sc.read_h5ad("ATAC_analysis/integrated_GEX_RNA.h5ad")
adata_tumor.obs['condition'] = "Tumor"

```

```

[25]: adata_schwann_spinal.obs

```

```

[25]:
      name cell_type leiden condition
AAAGGGCTCTTCTTCC-1-Spinal2  Spinal2  Schwann    15 Normal_Schwann_Spine
AACTTCTTCGGAGTAG-1-Spinal2  Spinal2  Schwann    18 Normal_Schwann_Spine
ACACCAAAGCGGGTAT-1-Spinal2  Spinal2  Schwann    15 Normal_Schwann_Spine
ACAGAAAGTATTGACC-1-Spinal2  Spinal2  Schwann    18 Normal_Schwann_Spine
ACGGGTCTCGACATTG-1-Spinal2  Spinal2  Schwann     5 Normal_Schwann_Spine
...
TGCCGAGCATGATGCT-1-Spinal10 Spinal10  Schwann     6 Normal_Schwann_Spine
TGTGCGGTCTGTACC-1-Spinal10 Spinal10  Schwann    15 Normal_Schwann_Spine
TTGCCTGAGGACAACC-1-Spinal10 Spinal10  Schwann    15 Normal_Schwann_Spine
TTGTTTGTGTCAGCATTG-1-Spinal10 Spinal10  Schwann    15 Normal_Schwann_Spine
TTTCACAAGTCCGCGT-1-Spinal10 Spinal10  Schwann    15 Normal_Schwann_Spine

```

```

[1289 rows x 4 columns]

```

```

[17]: adata_tumor = adata_tumor[adata_tumor.obs['pred.type'] == 'Schwann']
adata_tumor

```

```

[17]: View of AnnData object with n_obs × n_vars = 49639 × 38224
      obs: 'orig.ident', 'nCount_RNA', 'nFeature_RNA', 'type', 'percent.mt',
      'pANN_0.25_0.09_0', 'DF.classifications_0.25_0.09_0', 'RNA_snn_res.0.5',
      'seurat_clusters', 'pred.type', 'score1', 'X1p', 'X1q', 'X2p', 'X2q', 'X3p',
      'X3q', 'X4p', 'X4q', 'X5p', 'X5q', 'X6p', 'X6q', 'X7p', 'X7q', 'X8p', 'X8q',
      'X9p', 'X9q', 'X10p', 'X10q', 'X11p', 'X11q', 'X12p', 'X12q', 'X13q', 'X14q',

```

```
'X15q', 'X16p', 'X16q', 'X17p', 'X17q', 'X18q', 'X19p', 'X19q', 'X20p', 'X20q',
'X21q', 'X22q', 'x22q', 'integrated_snn_res.0.5', 'secondary_classification',
'Hippo1', 'score21', 'condition'
var: 'features'
obsm: 'X_umap'
```

```
[18]: def NF2_status(row):
        if row['score21'] >= 0.4403645:

            return "NF2_depleted"

        elif row['score21'] <= 0.1468272:

            return "NF2_intact"

        else:

            return "NF2_indeterminate"

adata_tumor.obs['NF2_status'] = adata_tumor.obs.apply(NF2_status, axis=1)
```

```
/lscratch/61866097/ipykernel_2383080/3528514657.py:14:
ImplicitModificationWarning: Trying to modify attribute `.obs` of view,
initializing view as actual.
adata_tumor.obs['NF2_status'] = adata_tumor.obs.apply(NF2_status, axis=1)
```

```
[19]: adata = [adata_tumor, adata_schwann_spinal, adata_schwann_nerve,
↳ adata_schwann_donor]
```

```
[20]: adata = sc.concat(adata, join='outer')
```

```
[26]: def sample_name(row):
        if row['condition'] == 'Tumor':

            return row['orig.ident']

        else:

            return row['name']

adata.obs['sample_name'] = adata.obs.apply(sample_name, axis=1)
```

```
[29]: sc.pp.pca(adata)
```

```
[30]: sc.external.pp.bbknnc(adata, batch_key='sample_name')
```

```
WARNING: consider updating your call to make use of `computation`
/data/bhatttda/conda/envs/palantir/lib/python3.8/site-packages/tqdm/auto.py:21:
```

TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm

```
[31]: sc.tl.umap(adata)
```

```
[41]: def NF2_status_label(row):  
        if row['condition'] == 'Tumor':  
  
            return row['NF2_status']  
  
        else:  
  
            return row['condition']  
  
adata.obs['NF2_status'] = adata.obs.apply(NF2_status_label, axis=1)  
adata.obs['NF2_status']
```

```
[41]: AAAAGTGAAGCTGCGAA-1_1      NF2_intact  
      AACAAACCAGTTAACGA-1_1      NF2_intact  
      AACACACCAGAACGCA-1_1      NF2_intact  
      AACCAACTCTTCCAGC-1_1      NF2_intact  
      AACCCAACACGTGAGA-1_1      NF2_intact  
  
      ...  
      GTTGCGGCATCAGCAT-1_Donor7  Normal_Schwann_Donor  
      TAAGTCGTCTGTCCCA-1_Donor7  Normal_Schwann_Donor  
      TCAGGGCTCGGCCAAC-1_Donor7  Normal_Schwann_Donor  
      TCTTCCTAGCTCAGAG-1_Donor7  Normal_Schwann_Donor  
      TTACGTTTCCTGCTAC-1_Donor7  Normal_Schwann_Donor  
Name: NF2_status, Length: 52034, dtype: object
```

```
[42]: fig, ax1 = plt.subplots(1,1,figsize=(6,6), dpi=600)  
ax1_dict = sc.pl.embedding(adata,  
                           basis="X_umap",  
                           use_raw = False,  
                           frameon=False,  
                           show=False,  
                           ax = ax1,  
                           color = "NF2_status",  
                           s = 20,  
                           palette = color_dict,  
                           edgecolor = 'black',  
                           linewidth= 0.2,  
                           alpha = 0.75,  
                           )
```

#add legend to plot

```

handles, labels = plt.gca().get_legend_handles_labels()

leg1 = ax1.legend(prop = {'size': '14'}, markerscale=1.5, bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.75)
    lh.set_edgecolor("black")

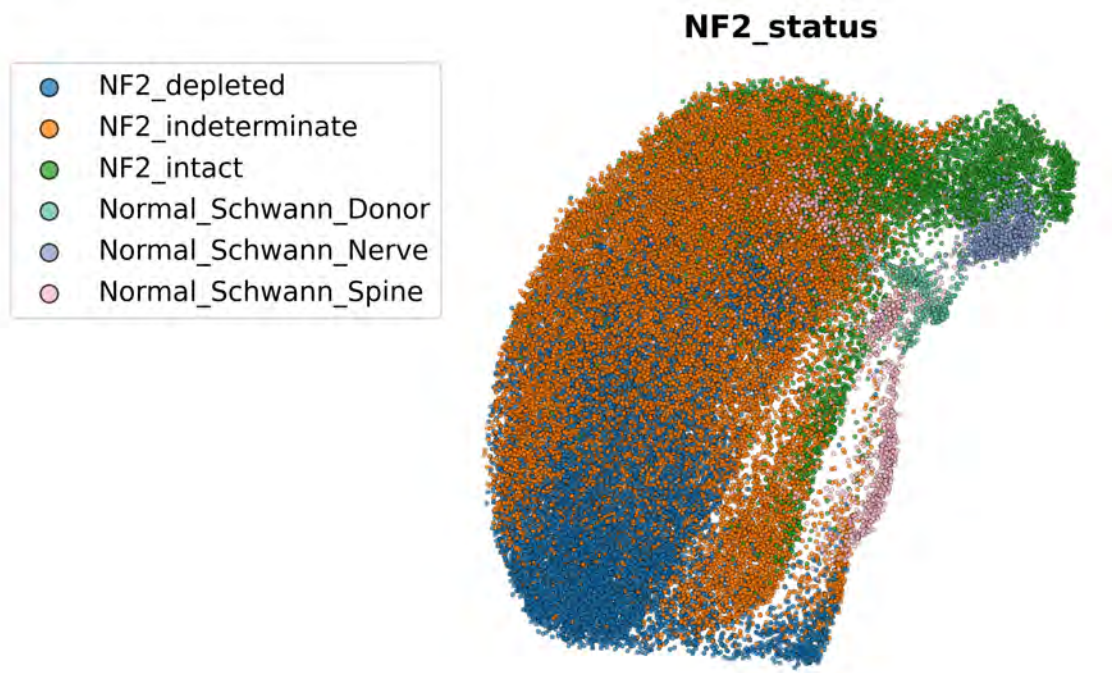
ax1.set_title("NF2_status", fontweight="bold", fontsize=16)
#ax2.set_title("Cluster", fontweight="bold", fontsize=16)
#ax3.set_title("Cell Type", fontweight="bold", fontsize=16)

fig.savefig('Schwan_NF2_status_with_Normal_Umap.png', bbox_inches='tight')

```

/data/bhatttda/conda/envs/palantir/lib/python3.8/site-packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for colormapping provided via 'c'. Parameters 'cmap' will be ignored

```
cax = scatter(
```



```

[69]: adata.obs['secondary_classification'] = adata_tumor.
      ↪obs['secondary_classification']
def subtype_label(row):
    if row['condition'] == 'Tumor':

        return row['secondary_classification']

```

```
else:
```

```
    return row['condition']
```

```
adata.obs['secondary_classification'] = adata.obs.apply(subtype_label, axis=1)
adata.obs['secondary_classification']
```

```
[69]: AAAGTGAAGCTGCGAA-1_1      Schwann_Stress
      AACAACCAGTTAACGA-1_1      Schwann_Macrophage
      AACACACCAGAACGCA-1_1      Schwann_Macrophage
      AACCAACTCTTCCAGC-1_1      Schwann_Macrophage
      AACCCAACACGTGAGA-1_1      Schwann_Classical
      ...
      GTTGCGGCATCAGCAT-1_Donor7 Normal_Schwann_Donor
      TAAGTCGTCTGTCCCA-1_Donor7 Normal_Schwann_Donor
      TCAGGGCTCGGCCAAC-1_Donor7 Normal_Schwann_Donor
      TCTTCCTAGCTCAGAG-1_Donor7 Normal_Schwann_Donor
      TTACGTTTCCTGCTAC-1_Donor7 Normal_Schwann_Donor
      Name: secondary_classification, Length: 52034, dtype: object
```

```
[ ]: genes_nf2_score = [
      "DAAM1", "ADARB1", "SAMD12", "NRG1", "COL12A1", "MIR4435-2HG", "CSNK1G3",
      ↪ "LYST", "AKAP6", "SPAG16",
      "TBC1D32", "LINC00968", "CHN1", "GPATCH2", "MIR222HG", "FAM126A", "AC012349.
      ↪ 1", "ROR1", "PGAP1", "CHIC2",
      "FSD1L", "FBXL2", "INTU", "AC096711.2", "PLSCR4", "TMEM168", "CMSS1",
      ↪ "COL5A2", "NEBL", "PPP2R2A", "ULK4",
      "GRB14", "ALDH1A2", "OLA1", "AL139383.1", "ELP4", "RANBP17", "GALNT18",
      ↪ "PREP", "TMEM241", "PDE5A",
      "GLIS3", "MPDZ", "CSTF3", "MINDY3", "PROS1", "SLC35F5", "DLG1", "ITGA1",
      ↪ "TBCK", "MAPK10", "IL1RAP",
      "FRS2", "GDAP1", "PSPC1", "ZDHHC21", "GALK2", "ZDHHC20", "NCKAP1", "NEK11",
      ↪ "MAOB", "ZNF248", "CUBN",
      "MEG8", "FRMD5", "ERBIN", "CDH2", "ANTXR2", "DNAJC6", "FBXL5", "AHCYL2",
      ↪ "DNAJC1", "LIMCH1", "NBAS",
      "TRAF3IP2-AS1", "VPS13A", "ANKRD13C", "DNAJB14", "COL24A1", "UBE3A",
      ↪ "NBEAL1", "TMEM260", "MAP3K20",
      "RHOBTB3", "CNKSR2", "USP45", "MEG3", "TANC1", "TC2N", "ZNF75D", "MAP2",
      ↪ "EPHB1", "CMC1", "PCDH9",
      "HERC4", "RAP1GDS1", "EPHA5-AS1", "DCAF10", "FBXO22", "UBR1", "WDR70",
      ↪ "SMYD3", "LINC00472", "FBXL4",
      "ITGB3BP", "TMEM267", "PKN2-AS1", "FAM184A", "MAP2K4", "ANKS1B", "FGF2",
      ↪ "TMTC2", "LRR6", "MRPL1",
      "CDC42BPA", "LANCL2", "KLHL13", "FHOD3", "WDFY3", "SCFD1", "SH3D19",
      ↪ "ZNF138", "WWP1", "RGS7", "PPME1",
```

```

        "P3H2", "DGKI", "DNM3", "NAV3", "DLG2", "C2CD3", "NPAS3", "SOS1", "LPAR1",
        ↪ "SIPA1L1", "GPHN", "NR2F1-AS1",
        "AP1S3", "FYN", "AC097460.1", "VPS13B", "AKT3", "DIAPH2", "SLC13A3", "EED",
        ↪ "ZC3H12C", "SLIT2", "CERS6",
        "NIPAL2", "MAGI2", "PTPN12", "TSPAN5", "EPHA3", "SUPT3H", "FAM151B",
        ↪ "USP53", "CRMP1", "FER"
    ]
    len(genes_nf2_score)

```

[]: 158

```

[56]: sc.tl.score_genes(adata, genes_nf2_score, score_name='NF2_depletion', use_raw=
        ↪ False)

```

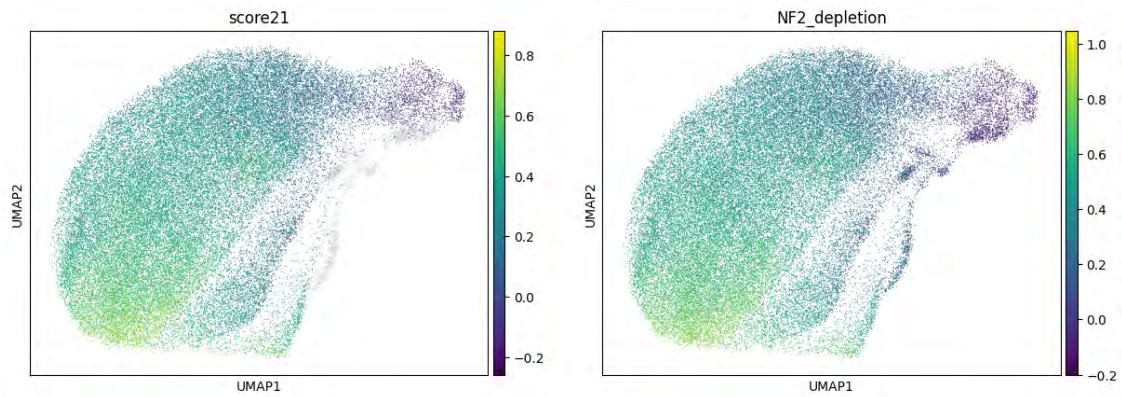
```

[67]: sc.pl.umap(adata, color = ['score21', 'NF2_depletion'])
fig, ax1 = plt.subplots(1,1,figsize=(6,6), dpi=600)
ax1_dict = sc.pl.embedding(adata,
                            basis="X_umap",
                            use_raw = False,
                            frameon=False,
                            show=False,
                            ax = ax1,
                            color = "NF2_depletion",
                            s = 20,
                            #palette = color_dict,
                            cmap = 'viridis',
                            edgecolor = 'black',
                            linewidth= 0.2,
                            alpha = 0.75,
                            )

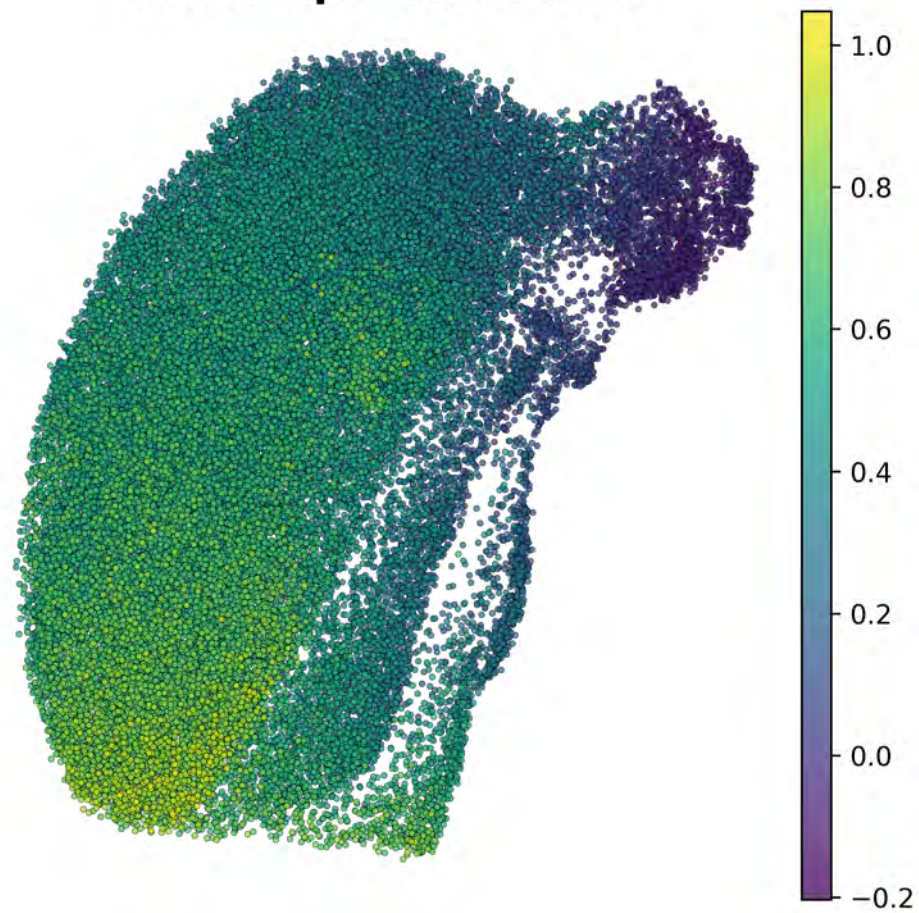
ax1.set_title("NF2 Depletion Score", fontweight="bold", fontsize=16)
#ax2.set_title("Cluster", fontweight="bold", fontsize=16)
#ax3.set_title("Cell Type", fontweight="bold", fontsize=16)

fig.savefig('Schwan_NF2_depletion_with_Normal_Umap.png', bbox_inches='tight')

```



NF2 Depletion Score



```
[58]: def NF2_status_label(row):
      if row['condition'] == 'Tumor':
```

```

        return row['NF2_status']

    else:

        return 'Normal'

adata.obs['NF2_status_Normal'] = adata.obs.apply(NF2_status_label, axis=1)
adata.obs['NF2_status_Normal']

```

```

[58]: AAAGTGAAGCTGCGAA-1_1      NF2_intact
      AACAACCAGTTAACGA-1_1      NF2_intact
      AACACACCAGAACGCA-1_1      NF2_intact
      AACCAACTCTTCCAGC-1_1      NF2_intact
      AACCCAACACGTGAGA-1_1      NF2_intact

      ...
      GTTGGGCGCATCAGCAT-1_Donor7    Normal
      TAAGTCGTCTGTCCCA-1_Donor7    Normal
      TCAGGGCTCGGCCAAC-1_Donor7    Normal
      TCTTCCTAGCTCAGAG-1_Donor7    Normal
      TTACGTTTCCTGCTAC-1_Donor7    Normal
Name: NF2_status_Normal, Length: 52034, dtype: object

```

```

[65]: sc.pl.violin(adata, keys='NF2_depletion', groupby = 'NF2_status_Normal',
    ↪rotation = 90,
        stripplot=False, # remove the internal dots
        inner="box",
        save= "_NF2_status_Normal_NF2_depletion_score.pdf"
    )

```

WARNING: saving figure to file
 figures/violin_NF2_status_Normal_NF2_depletion_score.pdf
 /data/bhattda/conda/envs/palantir/lib/python3.8/site-
 packages/scanpy/plotting/_anndata.py:839: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```

ax = sns.violinplot(
  /data/bhattda/conda/envs/palantir/lib/python3.8/site-
  packages/scanpy/plotting/_anndata.py:839: FutureWarning:

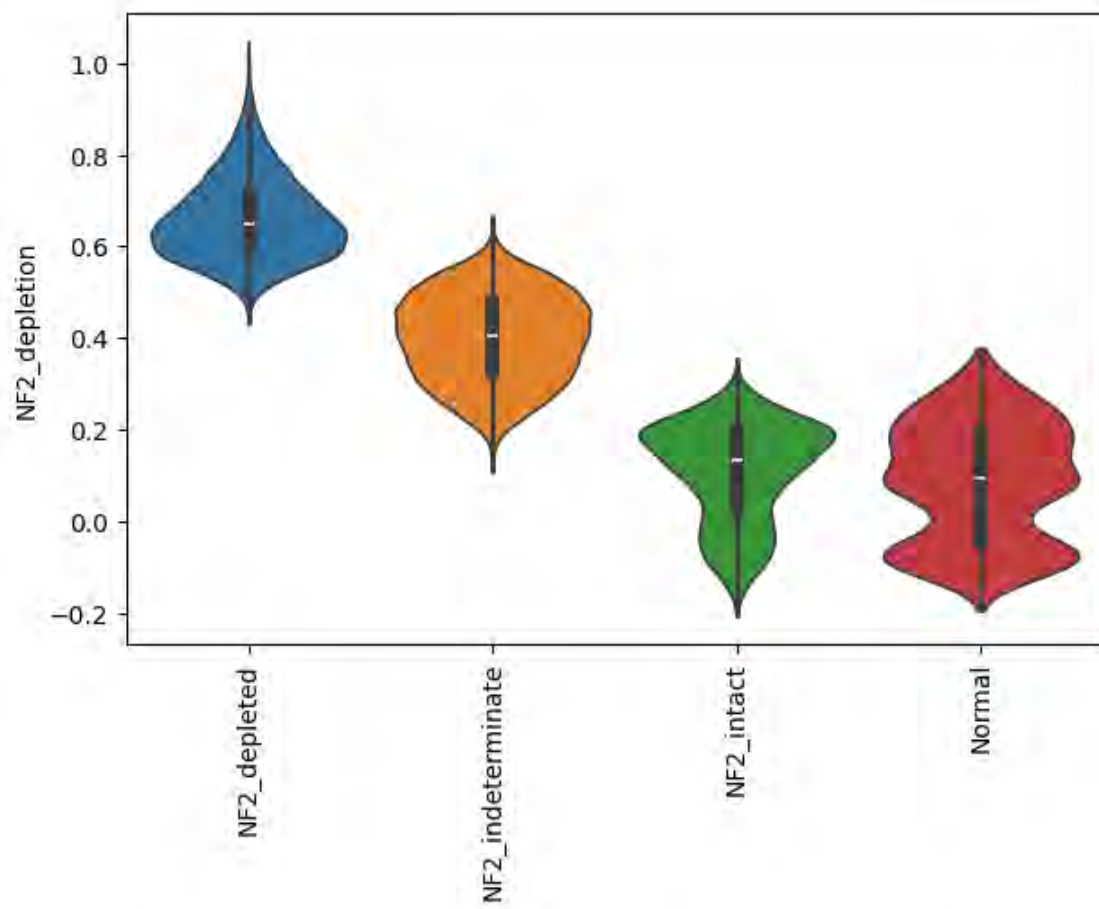
```

The `scale` parameter has been renamed and will be removed in v0.15.0. Pass `density\_norm='width'` for the same effect.

```

ax = sns.violinplot(

```



[]: